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Department of Computer Science and Engineering

Devops LAB

(CS505PC)

Lab Manual

EXPERIMENT NO: 1.

AIM:

Write code for a simple user registration form for an event.

DESCRIPTION:

Here's an example of a simple user registration form using Flask and Docker in DevOps:

- Create a Docker file with the following content to create a Docker image for your Flask application:

```
FROM python:3.8
WORKDIR /app
COPY . .
RUN pip install --no-cache-dir -r requirements.txt
EXPOSE 5000
CMD ["python", "app.py"]
```

- Create a requirements.txt file with the following content to list the dependencies of your Flask application: Flask==1.1.2
- Create a app.py file with the following code for a simple user registration form in Flask:

```
from flask import Flask, request, render_template
app = Flask(_name_)
@app.route('/register', methods=['GET', 'POST'])
def register():
    if request.method == 'POST':
        name = request.form['name']
        email = request.form['email']
        password = request.form['password']
        # Store the user data in a database or file
```

```
        return render_template('success.html')
    return render_template('register.html')
if__name__== '_main_':
    app.run(host='0.0.0.0')
```

- Create an templates folder and add the following two files: register.html and success.html.

register.html

```
<form method="post">
  <input type="text" name="name" placeholder="Name">
  <input type="email" name="email" placeholder="Email">
  <input type="password" name="password" placeholder="Password">
  <input type="submit" value="Submit">
</form>
```

success.html

```
<h2>Registration Successful</h2>
```

- Build the Docker image for your Flask application using the following command: docker build -t simple-flask-app .
- Run a Docker container from the image using the following command: docker run -p 5000:5000 simple-flask-app
- Open a web browser and access the registration form at http://localhost:5000/register.

This example demonstrates how to build a simple user registration form in Flask and run it in a Docker container in DevOps. Note that this code is only meant to demonstrate the basic structure of a user registration form and does not include any security measures or proper error handling. It is highly recommended to add security measures such as password hashing and validation before using it in a production environment.

EXPERIMENT NO: 2.

AIM:

Explore Git and GitHub commands

DESCRIPTION:

Git and GitHub are two of the most popular tools used for version control and collaboration in software development.

Here are some common Git and GitHub commands:

- Initializing a Git repository: $ git init
- Checking the status of your repository: $ git status
- Adding files to the stage: $ git add <file-name>
- Committing changes: $ git commit -m "commit message"
- Checking the commit history: $ git log
- Undoing changes: $ git checkout <file-name>
- Creating a new branch: $ git branch <branch-name>
- Switching to a different branch: $ git checkout <branch-name>
- Merging two branches: $ git merge <branch-name>
- Pushing changes to a remote repository: $ git push origin <branch-name>
- Cloning a repository from GitHub: $ git clone <repository-url>
- Creating a pull request on GitHub: Go to the repository on GitHub, select the branch you want to merge and click the "New pull request" button.

These are just a few of the many Git and GitHub commands available. There are many other Git commands and functionalities that you can explore to suit your needs.

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EXPERIMENT NO: 3.

AIM:

Practice Source code management on GitHub.Experiment with the source code written in exercise 1

DESCRIPTION:

To practice source code management on GitHub, you can follow these steps:

- Create a GitHub account if you don't already have one.
- Create a new repository on GitHub.
- Clone the repository to your local machine: $ git clone <repository-url>
- Move to the repository directory: $ cd <repository-name>
- Create a new file in the repository and add the source code written in exercise 1.
- Stage the changes: $ git add <file-name>
- Commit the changes: $ git commit -m "Added source code for a simple user registration form"
- Push the changes to the remote repository: $ git push origin master
- Verify that the changes are reflected in the repository on GitHub.

These steps demonstrate how to use GitHub for source code management. You can use the same steps to manage any source code projects on GitHub. Additionally, you can also explore GitHub features such as pull requests, code review, and branch management to enhance your source code management workflow.

EXPERIMENT NO: 4.

AIM:

Jenkins installation and setup, explore the environment

DESCRIPTION

Jenkins is a popular open-source tool for Continuous Integration and Continuous Deployment (CI/CD) in software development. Here are the steps to install and set up Jenkins:

Download and install Jenkins:

- Download the Jenkins package for your operating system from the Jenkins website.
- Follow the installation instructions for your operating system to install Jenkins.

Start the Jenkins service:

- On Windows, use the Windows Services Manager to start the Jenkins service.
- On Linux, use the following command to start the Jenkins service:
  $ sudo service jenkins start

Access the Jenkins web interface:

- Open a web browser and navigate to http://localhost:8080 to access the Jenkins web interface.
- If the Jenkins service is running, you will see the Jenkins login page.

Initialize the Jenkins environment:

- Follow the instructions on the Jenkins setup wizard to initialize the Jenkins environment.

- This process involves installing recommended plugins, setting up security, and creating the first admin user.

Explore the Jenkins environment:

- Once the Jenkins environment is set up, you can explore the various features and functionalities available in the web interface.
- Jenkins has a rich user interface that provides access to features such as build history, build statistics, and system information.

These are the basic steps to install and set up Jenkins. Depending on your use case, you may need to customize your Jenkins environment further. For example, you may need to configure build agents, set up build pipelines, or integrate with other tools. However, these steps should give you a good starting point for using Jenkins for CI/CD in your software development projects.

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EXPERIMENT NO: 5.

AIM:

Demonstrate continuous integration and development usingJenkins.

DESCRIPTION

Continuous Integration (CI) and Continuous Development (CD) are important practices in software development that can be achieved using Jenkins. Here's an example of how you can demonstrate CI/CD using Jenkins:

Create a simple Java application:

- Create a simple Java application that you want to integrate with Jenkins.
- The application should have some basic functionality, such as printing "Hello World" or performing simple calculations.

Commit the code to a Git repository:

- Create a Git repository for the application and commit the code to the repository.
- Make sure that the Git repository is accessible from the Jenkins server.

Create a Jenkins job:

- Log in to the Jenkins web interface and create a new job.
- Configure the job to build the Java application from the Git repository.
- Specify the build triggers, such as building after every commit to the repository.

Build the application:

- Trigger a build of the application using the Jenkins job.
- The build should compile the code, run any tests, and produce an executable jar file.

Monitor the build:

- Monitor the build progress in the Jenkins web interface.
- The build should show the build log, test results, and the status of the build.

Deploy the application:

- If the build is successful, configure the Jenkins job to deploy the application to a production environment.
- The deployment could be as simple as copying the jar file to a production server or using a more sophisticated deployment process, such as using a containerization technology like Docker.

Repeat the process:

- Repeat the process for subsequent changes to the application.
- Jenkins should automatically build and deploy the changes to the production environment.

This is a basic example of how you can use Jenkins to demonstrate CI/CD in software development. In a real-world scenario, you would likely have more complex requirements, such as multiple environments, different types of tests, and a more sophisticated deployment process. However, this example should give you a good starting point for using Jenkins for CI/CD in your software development projects.

EXPERIMENT NO. : 6.

AIM:

Explore Docker commands for content management.

DESCRIPTION

Docker is a containerization technology that is widely used for managing application containers. Here are some commonly used Docker commands for content management:

- Docker run: Run a command in a new container.

For example: $ docker run --name mycontainer -it ubuntu:16.04 /bin/bash

This command runs a new container based on the Ubuntu 16.04image and starts a shell session in the container.

- Docker start: Start one or more stopped containers.

For example: $ docker start mycontainer

This command starts the container named "mycontainer".

- Docker stop: Stop one or more running containers.

For example: $ docker stop mycontainer

This command stops the container named "mycontainer".

- Docker rm: Remove one or more containers.

For example: $ docker rm mycontainer

This command removes the container named "mycontainer".

- Docker ps: List containers.

For example: $ docker ps

This command lists all running containers.

- Docker images: List images.

For example: $ docker images

This command lists all images stored locally on the host.

- Docker pull: Pull an image or a repository from a registry.

For example: $ docker pull ubuntu:16.04

This command pulls the Ubuntu 16.04 image from the Docker Hub registry.

- Docker push: Push an image or a repository to a registry.

For example: $ docker push myimage

This command pushes the image named "myimage" to the Docker Hub registry.

These are some of the basic Docker commands for managing containers and images. There are many other Docker commands and options that you can use for more advanced use cases, such as managing networks, volumes, and configuration. However, these commands should give you a good starting point for using Docker for content management.

EXPERIMENT NO.: 7.

AIM:

Develop a simple containerized application using Docker

DESCRIPTION

Here's an example of how you can develop a simple containerized application using Docker:

Choose an application:

- Choose a simple application that you want to containerize. For example, a Python script that prints "Hello World".

Write a Dockerfile:

- Create a file named "Dockerfile" in the same directory as the application.

In the Dockerfile, specify the base image, copy the application into the container, and specify the command to run the application. Here's an example Dockerfile for a Python script:

```
# Use the official Python image as the base image
FROM python:3.9

# Copy the Python script into the container
COPY hello.py /app/

# Set the working directory to /app/
WORKDIR /app/
```

Run the Python script when the container starts
CMD ["python", "hello.py"]

- Build the Docker image:

Run the following command to build the Docker image:
$ docker build -t myimage .
This command builds a new Docker image using the Dockerfile and tags the image with the name "myimage".

- Run the Docker container:

Run the following command to start a new container based on the image:
$ docker run --name mycontainer myimage
This command starts a new container named "mycontainer" based on the "myimage" image and runs the Python script inside the container.

- Verify the output:

Run the following command to verify the output of the container:
$ docker logs mycontainer
This command displays the logs of the container and should show the "Hello World" output.

This is a simple example of how you can use Docker to containerize an application. In a real-world scenario, you would likely have more complex requirements, such as running multiple containers, managing network connections, and persisting data. However, this example should give you a good starting point for using Docker to containerize your applications.

EXPERIMENT NO.: 8.

AIM:

To Integrate Kubernetes and Docker

DESCRIPTION:

Kubernetes and Docker are both popular technologies for managing containers, but they are used for different purposes. Kubernetes is an orchestration platform that provides a higher-level abstraction for managing containers, while Docker is a containerization technology that provides a lower-level runtime for containers.

To integrate Kubernetes and Docker, you need to use Docker to build and package your application as a container image, and then use Kubernetes to manage and orchestrate the containers.

Here's a high-level overview of the steps to integrate Kubernetes and Docker:

- Build a Docker image:

Use Docker to build a Docker image of your application. You can use a Dockerfile to specify the base image, copy the application into the container, and specify the command to run the application.

- Push the Docker image to a registry:

Push the Docker image to a container registry, such as Docker Hub or Google Container Registry, so that it can be easily accessed by Kubernetes.

Deploy the Docker image to a Kubernetes cluster:

Use Kubernetes to deploy the Docker image to a cluster. This involves

creating a deployment that specifies the number of replicas and the image to be used, and creating a service that exposes the deployment to the network.
Monitor and manage the containers:

Use Kubernetes to monitor and manage the containers. This includes scaling the number of replicas, updating the image, and rolling out updates to the containers.

- Continuously integrate and deploy changes:

Use a continuous integration and deployment (CI/CD) pipeline to automatically build, push, and deploy changes to the Docker image and the Kubernetes cluster. This makes it easier to make updates to the application and ensures that the latest version is always running in the cluster.
By integrating Kubernetes and Docker, you can leverage the strengths of both technologies to manage containers in a scalable, reliable, and efficient manner.

Viva questions:

Define integrate Kubernetes
What is Docker

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EXPERIMENT NO.: 9.

AIM:

Automate the process of running containerized applicationdeveloped in exercise 7 using Kubernetes

DESCRIPTION

To automate the process of running the containerized application developed in exercise 7 using Kubernetes, you can follow these steps:

- Create a Kubernetes cluster:

Create a Kubernetes cluster using a cloud provider, such as Google Cloud or Amazon Web Services, or using a local installation of Minikube.

- Push the Docker image to a registry:

Push the Docker image of your application to a container registry, such as Docker Hub or Google Container Registry.

- Create a deployment:

Create a deployment in Kubernetes that specifies the number of replicas and the Docker image to use. Here's an example of a deployment YAML file:

```
apiVersion: apps/v1
kind: Deployment
metadata:

  name: myapp
spec:
  replicas: 3
```

```
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
    spec:
      containers:
      - name: myapp
        image: myimage
        ports:
        - containerPort: 80
```

- Create a service:

Create a service in Kubernetes that exposes the deployment to the network. Here's an example of a service YAML file:

```
apiVersion: v1
kind: Service
metadata:
  name: myapp-service
spec:
  selector:
    app: myapp
  ports:
  - name: http
    port: 80
    targetPort: 80
```

type: ClusterIP

- Apply the deployment and service to the cluster:

Apply the deployment and service to the cluster using the kubectl command-line tool. For example:

$ kubectl apply -f deployment.yaml

$ kubectl apply -f service.yaml

- Verify the deployment:

Verify the deployment by checking the status of the pods and the service. For example:

$ kubectl get pods

$ kubectl get services

This is a basic example of how to automate the process of running a containerized application using Kubernetes. In a real-world scenario, you would likely have more complex requirements, such as managing persistent data, scaling, and rolling updates, but this example should give you a good starting point for using Kubernetes to manage your containers.

VIVA QUESTIONS

Define Kubernetes

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EXPERIMENT NO.: 10.

AIM:

Install and Explore Selenium for automated testing

DESCRIPTION:

To install and explore Selenium for automated testing, you can follow these steps:

Install Java Development Kit (JDK):

- Selenium is written in Java, so you'll need to install JDK in order to run it. You can download and install JDK from the official Oracle website.
- Install the Selenium WebDriver:
- You can download the latest version of the Selenium WebDriver from the Selenium website. You'll also need to download the appropriate driver for your web browser of choice (e.g. Chrome Driver for Google Chrome).

Install an Integrated Development Environment (IDE):

- To write and run Selenium tests, you'll need an IDE. Some popular choices include Eclipse, IntelliJ IDEA, and Visual Studio Code.
- Write a simple test:
- Once you have your IDE set up, you can write a simple test using the Selenium WebDriver. Here's an example in Java:

```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class Main {
  public static void main(String[] args) {
    System.setProperty("webdriver.chrome.driver", "path/to/chromedriver");
    WebDriver driver = new ChromeDriver();
    driver.get("https://www.google.com");
    System.out.println(driver.getTitle());
    driver.quit();
  }
}
```

- Run the test:

Run the test using your IDE or from the command line using the following command:

```
$ javac Main.java
$ java Main
```

This is a basic example of how to get started with Selenium for automated testing. In a real-world scenario, you would likely write more complex tests and organize your code into test suites and test cases, but this example should give you a good starting point for exploring Selenium.

EXPERIMENT NO.: 11.

AIM:

Write a simple program in JavaScript and perform testing usingSelenium

PROGRAM:

Simple JavaScript program that you can test using Selenium

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple JavaScript Program</title>
</head>
<body>
  <p id="output">0</p>
  <button id="increment-button">Increment</button>
  <script>
        const output = document.getElementById("output");
        const incrementButton = document.getElementById("increment-button");

        let count = 0;
        incrementButton.addEventListener("click", function() {
        count += 1;
        output.innerHTML = count;
        });
  </script>
```

```
</body>
</html>
```

Write a test case for this program using Selenium

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.junit.After;
import org.junit.Before;
import org.junit.Test;

public class Main {
  private WebDriver driver;

  @Before
  public void setUp() {
    System.setProperty("webdriver.chrome.driver", "path/to/chromedriver");
    driver = new ChromeDriver();
  }

  @Test
  public void testIncrementButton() {
    driver.get("file:///path/to/program.html");
    driver.findElement(By.id("increment-button")).click();
    String result = driver.findElement(By.id("output")).getText();
    assert result.equals("1");
  }

  @After
  public void tearDown() {
```

```
    driver.quit();
  }
}
```

You can run the test case using the following command:

$ javac Main.java

$ java Main

The output of the test case should be:

.

Time: 0.189

OK (1 test)

This output indicates that the test case passed, and the increment button was successfully clicked, causing the output to be incremented by 1.

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EXPERIMENT NO.: 12.

AIM:

Develop test cases for the above containerized application usingselenium

PROGRAM:

Here is an example of how you could write test cases for the containerizedapplication using Selenium

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.junit.After;
import org.junit.Before;
import org.junit.Test;

public class Main {
  private WebDriver driver;

  @Before
  public void setUp() {
    System.setProperty("webdriver.chrome.driver", "path/to/chromedriver");
    driver = new ChromeDriver();
  }

  @Test
  public void testHomePageLoads() {
```

```
    driver.get("http://localhost:8080");
    String title = driver.getTitle();
    assert title.equals("My Containerized Application");
  }
  @Test
  public void testSubmitForm() {
    driver.get("http://localhost:8080");
    driver.findElement(By.name("name")).sendKeys("John Doe");
      driver.findElement(By.name("email")).sendKeys("john.doe@example.co
m");
    driver.findElement(By.name("submit")).click();
    String result = driver.findElement(By.id("result")).getText();
    assert result.equals("Form submitted successfully!");
  }
  @After
  public void tearDown() {
    driver.quit();
  }
}
```

You can run the test cases using the following command:

```
$ javac Main.java
$ java Main
```

The output of the test cases should be:

```
..
Time: 1.135

OK (2 tests)
```

This output indicates that both test cases passed, and the containerized

application is functioning as expected.